

Database Technologies – Assignment 5

1. Display all the first names in lower case and the last names in upper case as the single column separated with ' - '. Provide the meaningful name to the column.

```
KD1_Rosh_98811> SELECT CONCAT(LOWER(first_name), ' - ', UPPER(last_name)) AS employee_name
-> FROM employees;
+-----+
| employee_name |
+-----+
| ellen - ABEL  |
| sundar - ANDE |
| mozhe - ATKINSON |
| shelli - BAIDA |
| amit - BANDA  |
| elizabeth - BATES |
| sarah - BELL   |
| david - BERNSTEIN |
| laura - BISSOT  |
| harrison - BLOOM |
| hermann - BROWN |
| alexis - BULL   |
| anthony - CABRIO |
| gerald - CAMBRAULT |
| nanette - CAMBRAULT |
| john - CHEN     |
| kelly - CHUNG   |
| karen - COLMENARES |
| curtis - DAVIES |
| pat - DAVIS     |
| julia - DELLINGER |
| jennifer - DILLY |
| louise - DORAN  |
| alberto - ERRAZURIZ |
| britney - EVERETT |
| daniel - FAVIET  |
| kevin - FEENEY   |
| jean - FLEAUR   |
| taylor - FOX     |
| adam - FRIPP    |
| lex - GARCIA    |
| ki - GEE        |
| girard - GEONI   |
| william - GIETZ  |
| douglas - GRANT  |
+-----+
```

2. Display the first word of the job title. (Hint : Use substring_index).

```
KD1_Rosh_98811> SELECT SUBSTRING_INDEX(job_title, ' ', 1) AS first_word FROM jobs;
+-----+
| first_word |
+-----+
| Public    |
| Accounting |
| Administration |
| President |
| Administration |
| Accountant |
| Finance   |
| Human     |
| Programmer |
| Marketing |
| Marketing |
| Public    |
| Purchasing |
| Purchasing |
| Sales     |
| Sales     |
| Shipping  |
| Stock     |
| Stock     |
+-----+
19 rows in set (0.334 sec)
```

3. Display the First name, last name and length of first name for the employees whose last name contains character 'b' anywhere after 3rd position.

```
KD1_Rosh_98811> SELECT first_name, last_name, LENGTH(first_name) AS first_name_length FROM employees WHERE SUBSTRING(last_name, 4) LIKE '%b%';
```

first_name	last_name	first_name_length
Gerald	Cambrault	6
Nanette	Cambrault	7
Nancy	Gruenberg	5
Susan	Jacobs	5

4 rows in set (0.019 sec)

4. Display the employee id, first name and last name of the employees whose first or last name includes a space in between.

```
KD1_Rosh_98811> SELECT employee_id, first_name, last_name
-> FROM employees
-> WHERE first_name LIKE '% %' OR last_name LIKE '% %';
```

employee_id	first_name	last_name
112	Jose Manuel	Urman

1 row in set (0.007 sec)

5. Display first name, salary, and round the salary to thousands.

```
KD1_Rosh_98811> SELECT first_name, salary, ROUND(salary, -3) AS rounded_salary FROM employees;
```

first_name	salary	rounded_salary
Steven	24000.00	24000
Neena	17000.00	17000
Lex	17000.00	17000
Alexander	9000.00	9000
Bruce	6000.00	6000
David	4800.00	5000
Valli	4800.00	5000
Diana	4200.00	4000
Nancy	12008.00	12000
Daniel	9000.00	9000
John	8200.00	8000
Ismael	7700.00	8000
Jose Manuel	7800.00	8000
Luis	6900.00	7000
Den	11000.00	11000
Alexander	3100.00	3000
Shelli	2900.00	3000
Sigal	2800.00	3000
Guy	2600.00	3000
Karen	2500.00	3000
Matthew	8000.00	8000
Adam	8200.00	8000
Payam	7900.00	8000
Shanta	6500.00	7000
Kevin	5800.00	6000
Julia	3200.00	3000
Irene	2700.00	3000
James	2400.00	2000
Steven	2200.00	2000
Laura	3300.00	3000
Mozhe	2800.00	3000
James	2500.00	3000
TJ	2100.00	2000
Jason	3300.00	3000
Michael	2900.00	3000
Ki	2400.00	2000

6. Display employee ID and the date on which he ended his previous job in the format as mentioned. (Format: Thursday, October 27th 2011)

```

K01_Rosh_98811> SELECT employee_id, END_DATE, DATE_FORMAT(end_date, '%W, %M %D %Y') AS previous_job_end_date FROM job_history;

```

employee_id	END_DATE	previous_job_end_date
101	2011-10-27	Thursday, October 27th 2011
101	2015-03-15	Sunday, March 15th 2015
102	2016-07-24	Sunday, July 24th 2016
114	2017-12-31	Sunday, December 31st 2017
122	2017-12-31	Sunday, December 31st 2017
176	2016-12-31	Saturday, December 31st 2016
176	2017-12-31	Sunday, December 31st 2017
200	2011-06-17	Friday, June 17th 2011
200	2016-12-31	Saturday, December 31st 2016
201	2017-12-19	Tuesday, December 19th 2017

10 rows in set (0.006 sec)

7. Display first name and date of first salary of the employees. (Consider 1st day of month as salary day. HINT: LAST_DAY).

```

K01_Rosh_98811> SELECT first_name,
-> ADDDATE(LAST_DAY(hire_date), INTERVAL 1 DAY) AS first_salary_date
-> FROM employees;

```

first_name	first_salary_date
Steven	2013-07-01
Neena	2015-10-01
Lex	2011-02-01
Alexander	2016-02-01
Bruce	2017-06-01
David	2015-07-01
Valli	2016-03-01
Diana	2017-03-01
Nancy	2012-09-01
Daniel	2012-09-01
John	2015-10-01
Ismael	2015-10-01
Jose Manuel	2016-04-01
Luis	2018-01-01
Den	2013-01-01
Alexander	2013-06-01
Shelli	2016-01-01
Sigal	2015-08-01
Guy	2016-12-01
Karen	2017-09-01
Matthew	2014-08-01
Adam	2015-05-01
Payam	2013-06-01
Shanta	2015-11-01
Kevin	2017-12-01
Julia	2015-08-01
Irene	2016-10-01
James	2017-02-01
Steven	2018-04-01
Laura	2015-09-01
Mozhe	2015-11-01
James	2015-03-01
TJ	2017-05-01
Jason	2014-07-01

8. Display first name and experience of the employees in the format of number of years, months and days completed.

```

K01_Rosh_98811> SELECT first_name, hire_date,
-> CONCAT(TIMESTAMPDIFF(YEAR, hire_date, CURDATE()), ' years ',
-> TIMESTAMPDIFF(MONTH, DATE_ADD(hire_date, INTERVAL TIMESTAMPDIFF(YEAR, hire_date, CURDATE()) YEAR),
-> CURDATE()), ' months ',
-> DATEDIFF(CURDATE(),
-> DATE_ADD(DATE_ADD(hire_date, INTERVAL TIMESTAMPDIFF(YEAR, hire_date, CURDATE()) YEAR),
-> INTERVAL TIMESTAMPDIFF(MONTH, DATE_ADD(hire_date, INTERVAL TIMESTAMPDIFF(YEAR, hire_date, CURDATE()) YEAR), CURDATE()) MONTH)), ' days') AS
experience
-> FROM employees;

```

first_name	hire_date	experience
Steven	2013-06-17	13 years 2 months 21 days
Neena	2015-09-21	10 years 11 months 17 days
Lex	2011-01-13	15 years 7 months 25 days
Alexander	2016-01-03	10 years 8 months 4 days
Bruce	2017-05-21	9 years 3 months 17 days
David	2015-06-25	11 years 2 months 13 days
Valli	2016-02-05	10 years 7 months 2 days
Diana	2017-02-07	9 years 7 months 0 days
Nancy	2012-08-17	14 years 8 months 21 days
Daniel	2012-08-16	14 years 8 months 22 days
John	2015-09-28	10 years 11 months 10 days
Ismael	2015-09-30	10 years 11 months 8 days
Jose Manuel	2016-03-07	10 years 6 months 0 days
Luis	2017-12-07	8 years 9 months 0 days
Den	2012-12-07	13 years 9 months 0 days
Alexander	2013-05-18	13 years 3 months 20 days
Shelli	2015-12-24	10 years 8 months 14 days
Sigal	2015-07-24	11 years 1 months 14 days
Guy	2016-11-15	9 years 9 months 23 days
Karen	2017-08-10	9 years 0 months 20 days
Matthew	2014-07-18	12 years 1 months 20 days
Adam	2015-04-10	11 years 4 months 28 days
Payam	2013-05-01	13 years 4 months 6 days
Shanta	2015-10-10	10 years 10 months 28 days
Kevin	2017-11-16	8 years 9 months 22 days
Julia	2015-07-16	11 years 1 months 22 days
Irene	2016-09-28	9 years 11 months 10 days
James	2017-01-14	9 years 7 months 24 days

9. Display the employee id, first name and joining date of the employees who joined in the month of august of any year.

```
KD1_Rosh_98811> SELECT employee_id, first_name, hire_date FROM employees where month(hire_date)=8;
+-----+-----+-----+
| employee_id | first_name | hire_date |
+-----+-----+-----+
|          108 | Nancy      | 2012-08-17 |
|          109 | Daniel     | 2012-08-16 |
|          119 | Karen      | 2017-08-10 |
|          129 | Laura      | 2015-08-20 |
|          134 | Michael    | 2016-08-26 |
|          152 | Peter      | 2015-08-20 |
|          158 | Allan      | 2014-08-01 |
|          189 | Jennifer   | 2015-08-13 |
|          202 | Pat        | 2015-08-17 |
+-----+-----+-----+
9 rows in set (0.043 sec)
```

10. Display the details of employees who joined before the year 2015.

```
KD1_Rosh_98811> SELECT employee_id, hire_date
-> FROM employees
-> WHERE hire_date < '2015-01-01';
+-----+-----+
| employee_id | hire_date |
+-----+-----+
|          100 | 2013-06-17 |
|          102 | 2011-01-13 |
|          108 | 2012-08-17 |
|          109 | 2012-08-16 |
|          114 | 2012-12-07 |
|          115 | 2013-05-18 |
|          120 | 2014-07-18 |
|          122 | 2013-05-01 |
|          133 | 2014-06-14 |
|          137 | 2013-07-14 |
|          141 | 2013-10-17 |
|          145 | 2014-10-01 |
|          156 | 2014-01-30 |
|          157 | 2014-03-04 |
|          158 | 2014-08-01 |
|          174 | 2014-05-11 |
|          184 | 2014-01-27 |
|          192 | 2014-02-04 |
|          200 | 2013-09-17 |
|          201 | 2014-02-17 |
|          203 | 2012-06-07 |
|          204 | 2012-06-07 |
|          205 | 2012-06-07 |
|          206 | 2012-06-07 |
+-----+-----+
24 rows in set (0.006 sec)
```

11. Display the number of days between system date and 1st January 2011.

```
KD1_Rosh_98811> SELECT DATEDIFF(NOW(), '2011-01-01') AS number_of_days;
+-----+
| number_of_days |
+-----+
|          5728 |
+-----+
1 row in set (0.016 sec)
```

12. Display number of employees joined after 15th of any month.

```
KD1_Rosh_98811> SELECT COUNT(*) AS employee_count
-> FROM employees
-> WHERE DAYOFMONTH(hire_date) > 15;
+-----+
| employee_count |
+-----+
|          57 |
+-----+
1 row in set (0.030 sec)
```

13. Display third highest salary of employees.

```
KD1_Rosh_98811> SELECT salary FROM EMPLOYEES GROUP BY SALARY ORDER BY SALARY DESC LIMIT 2,1;
+-----+
| salary |
+-----+
| 14000.00 |
+-----+
1 row in set (0.058 sec)
```